

Supporting Information

Which modelling choices are supported when marine survey data are limited? An evidence map of 95 studies and a benchmark on structured subsamples of a trawl survey

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Appendix S1. Search strategy, screening, inclusion criteria, and non-marine sources retained

S1.1 Databases and coverage

Searches combined method terms (Random Forest, boosted regression trees, VAST, sdmTMB, generalised additive model, MaxEnt, joint species distribution model, graph neural network), spatial terms (spatial autocorrelation, spatial cross-validation, geostatistical, random field), data limitation terms (data-limited, small-sample, sparse) and marine ecosystem descriptors (marine, ocean, estuarine, fish, cetacean, benthic). Complete Boolean strings are in Table S2.

Table S1. Records identified, screened and included by source. Counts are unique records after duplicate removal within each source at each stage. Thirteen studies eligible at full-text were duplicates across sources and were resolved at the final stage, so the per source included counts sum to 108 before that resolution and 95 after it.

Source	Period	Identified	After title and abstract	After full-text	Included
Web of Science	2001 to 2025	185	89	58	50
Google Scholar	2001 to 2025	95	48	28	25
Discipline-specific journals	2010 to 2025	35	25	14	13
Reference tracking		15	10	8	8
<i>Total</i>		<i>330</i>	<i>172</i>	<i>108</i>	<i>95</i>

Table S2. Complete Boolean search strings by topic area. Strings were executed across Web of Science and Google Scholar; the discipline-specific journal search used the same topic terms applied to journal level search interfaces.

#	Topic area	Boolean search string	Rationale
1	General spatial ML and SDM	("spatial machine learning" OR "species distribution model*" OR "habitat model*" OR "niche model*" OR "SDM" OR "HSM") AND (marine OR ocean* OR aquatic OR fish* OR cetacean* OR "marine mammal*" OR benthos OR benthic OR pelagic OR demersal OR "marine ecosystem*")	Broad capture of spatial modelling approaches in marine systems
2	Random Forest	("random forest*" OR "RF" OR "ensemble learning" OR "random tree*") AND (marine OR ocean* OR fish* OR cetacean* OR aquatic) AND (spatial OR distribution OR habitat OR predict*)	Specific RF method with marine applications
3	Boosted Regression Trees	("boosted regression tree*" OR "BRT" OR "gradient boosting" OR "GBM" OR "boosted tree*" OR "gradient boost*") AND (marine OR ocean* OR fish* OR "marine species" OR cetacean*) AND (distribution OR habitat OR spatial)	BRT and gradient boosting variants
4	VAST framework	("VAST" OR "vector autoregressive spatio-temporal" OR "vector autoregressive spatiotemporal" OR "geostatistical delta" OR "spatio-temporal GLMM") AND (fisheries OR marine OR ocean* OR fish OR "stock assessment")	Purpose built marine spatio temporal framework
5	Spatial cross validation	("spatial cross-validation" OR "spatial cross validation" OR "spatial CV" OR "spatial autocorrelation" OR "spatial validation" OR "spatial blocking") AND ("species distribution" OR "habitat model*" OR SDM OR "predictive model*" OR "machine learning")	Critical validation methodology
6	GAM spatial models	("generalised additive model*" OR "GAM" OR "additive model*") AND spatial AND (marine OR ocean* OR fish* OR aquatic) AND (distribution OR habitat)	Traditional and hybrid spatial modelling approach
7	MaxEnt	("MaxEnt" OR "maximum entropy" OR "Maxent") AND (marine OR ocean* OR aquatic OR fish*) AND (species OR distribution OR habitat OR SDM)	Presence only modelling dominant method
8	Transfer learning	("transfer learning" OR "extrapolation" OR "transferability" OR "model transfer*" OR "spatial transfer*" OR "temporal transfer*") AND (species OR marine OR ecology OR distribution OR SDM) AND (predict* OR model*)	Transferability and forecast skill assessment
9	Data limited contexts	("data limited" OR "data-limited" OR "data poor" OR "limited sample*" OR "small sample*" OR "sparse data") AND (marine OR fisheries OR ocean*) AND (model* OR predict* OR distribution OR SDM)	Explicit focus on data constraints

Table S2 (continued)

#	Topic area	Boolean search string	Rationale
10	Graph neural networks	("graph neural network*" OR "GNN" OR "deep learning" OR "neural network*" OR "convolutional neural") AND (marine OR ocean* OR species OR ecology OR spatial) AND (distribution OR habitat OR predict*)	Emerging deep learning methods
11	Uncertainty quantification	("uncertainty quantification" OR "prediction uncertainty" OR "model uncertainty" OR "adaptive monitoring" OR "value of information") AND ("species distribution" OR marine OR ecology OR SDM) AND (spatial OR predict*)	Uncertainty and adaptive approaches
12	Spatial Random Forest specifically	("spatial random forest" OR "spatial RF" OR "RFsp" OR "geographically weighted random forest") AND (ecology OR species OR distribution OR marine OR spatial)	Enhanced spatial RF implementations
13	Ensemble methods	(ensemble OR "model averaging" OR "multi-model" OR "stacked model*") AND ("species distribution" OR SDM OR marine OR ecology) AND (predict* OR spatial)	Multiple model combinations
14	Validation reviews	("cross-validation" OR "model validation" OR "performance assessment") AND spatial AND (autocorrelation OR blocking OR structure) AND (ecology OR species OR SDM)	Methodological validation literature
15	Joint species distribution models	("joint species distribution" OR "JSDM" OR "multivariate species" OR "HMSC" OR "hierarchical modelling of species communities" OR "community-level model") AND (marine OR ocean* OR fish* OR aquatic) AND (spatial OR distribution OR predict*)	Joint modelling approaches leveraging data sharing

S1.2 Inclusion and exclusion criteria

Included: marine or estuarine application; a spatial modelling component; sample size reported; performance metrics or documented validation practice; peer-reviewed and in English.

Excluded: purely terrestrial systems, except where a study provided a methodological result with no marine equivalent, in which case it was retained and flagged; reviews without original analysis; insufficient methodological detail to determine the spatial treatment or validation design; no performance assessment of any kind.

Corpus membership and citation are separate. The corpus comprises studies that passed the criteria above and contributed evidence to the synthesis. The reference list additionally contains statistical, algorithmic and software sources that have no study system at all, and which are cited for provenance or implementation rather than as evidence; these were never screened as corpus candidates and are not counted in any total reported in this review. Table S3 lists every non-marine source cited, distinguishing those that anchor a finding from those cited for provenance only.

Table S3. Sources applied outside marine and estuarine systems, or to simulated data, cited in the main text. Each is identified as non-marine at every point of use, and the consequence for the evidence base is addressed in main text Sections 3.1 and 4.4. The classification basis records how the system was determined: verified against the publisher record, from the article title, or from the journal scope.

Study	System	Reason for retention	Finding supported	Basis
<i>Studies anchoring a finding (corpus, retained under the methodological exception)</i>				
Hengl et al. (2018)	Soil and environmental benchmarks, terrestrial	The only study in the review that crosses spatial treatment against algorithm family on identical data; its kriging comparison also prevents the spatial structure finding being read as an endorsement of tree ensembles	Spatial structure finding (main text 3.2); Table 5	Title
Talebi et al. (2022)	Geoscience	Independent corroboration of the same contrast in a second domain, reported on error rather than discrimination metrics	Spatial structure finding (3.2)	Journal
Mushagalusa et al. (2024)	Simulated abundance data; no study system	Compares standard, design-based, buffered, environmentally blocked and spatially blocked partitioning on data generated from a Gaussian process; simulation is the only setting in which the true surface is known, so validation optimism can be measured rather than inferred	Spatial structure and validation findings (3.2)	Verified
Ploton et al. (2020)	Tropical forest biomass, terrestrial	The reference demonstration that random partitioning inflates apparent accuracy under spatial autocorrelation; its 11.8 million observations have no marine counterpart	Validation finding (3.2)	Title
Mi et al. (2017)	Cranes, wetland	One of only two studies quantifying performance below 100 observations; no marine study in the corpus compares algorithms at comparable sample sizes	Coverage and minimum data findings (3.2, 3.4); Tables 2 and 6	Title

Table S3 (continued)

Study	System	Reason for retention	Finding supported	Basis
<i>Sources cited for method provenance or implementation (not corpus candidates)</i>				
Behrens et al. (2018)	Soil, terrestrial	Introduces Euclidean distance fields, the spatial feature construction on which spatial tree ensembles depend	Section 3.5 (coordinate representation)	Journal
Elith et al. (2006)	226 species across six regions, terrestrial	The benchmark comparison for presence-only methods; cited for provenance and as the data-rich regime this review distinguishes itself from	Introduction	Title
Norberg et al. (2019)	Multiple, primarily non-marine datasets	Comparison of 33 methods; cited for the joint model finding that spatial shrinkage variants outperform non-spatial joint models	Section 3.5; Tables 6 and 7	Verified
<i>Statistical and machine learning sources with no study system</i>				
Hastie et al. (2009)	Statistical theory	Rotation dependence of axis-aligned tree splits	Section 3.5 (coordinate representation)	Verified

Note for reconciliation: main text Section 2.1 states that eight non-marine studies were retained under the exception. Table S3 lists five that anchor a finding and three further non-marine sources cited for provenance; the count stated in the main text should be made to match the corpus screening records before the preprint is posted.

S1.3 How main text Table 1 was derived

The evidence map (main text Table 1) counts sources cited in support of each contrast, attributed from the citation commands in the main text. A source cited under more than one contrast is counted under each, so the columns do not sum to the number of distinct sources. These counts are a property of the argument as written and can be checked against the printed paper; they are a lower bound on corpus contribution, since a study may inform a finding without being cited where that finding is stated. Marine status follows Table S3.

The counts were derived from the per finding coding of the earlier version of this synthesis (Table S4) and then reassigned to the contrast rows of main text Table 1. Table S4 is retained so that the derivation can be checked.

Table S4. Distinct sources cited in support of each finding in the earlier version of this synthesis, counted from citation commands in that main text.

Finding	Sources cited	Of which marine	Of which crossing both contrasts
Spatial structure before algorithmic complexity	6	3	3
Coverage before sample size	6	4	0
Validation matched to inference target	4	3	0
Decision framework and applications	15	15	0
Method families (Table 2 of the main text)	21	17	1
Quantitative claims (Table 5 of the main text)	9	7	1

The pattern is the content of main text Section 3.1: support for the spatial structure finding rests on six sources, three of them marine, and all three studies crossing spatial treatment against algorithm family are non-marine; coverage and validation rest on small numbers of sources reporting the relevant experiment directly; the decision framework draws on fifteen marine sources because it synthesises applied work rather than resting on comparative experiments.

S1.4 Relation to previous reviews

Table S5. Contribution relative to previous reviews of marine species distribution modelling and machine learning in marine systems.

Review	Scope	Quantitative synthesis	Data limitation addressed	Validation guidance	Minimum data thresholds	Evidence map
Robinson et al. (2017)	Marine SDM practice	No	No	Partial	No	No
Melo-Merino et al. (2020)	ML in marine systems	Bibliometric	No	No	No	No
Rubbens et al. (2023)	ML in marine ecology	No	No	Partial	No	No
Pasanisi et al. (2024)	Marine SDM	No	No	No	No	No
Klaassen et al. (2025)	Methodological trends	No	No	No	No	No
This review	Marine SDM under data limitation	Directional, with attributed within-study values	Yes, as the organising scope	Yes, matched to inference target	Yes, method-specific and graded	Yes

Appendix S2. Extraction protocol, extracted studies and quality assessment

S2.1 Selection for quantitative extraction

Twelve studies were selected on the basis of at least one of four criteria: a direct method comparison on identical data; an explicit assessment of validation optimism; a sample size experiment with associated performance metrics; or an authoritative performance benchmark for a specific method. Selection was made before extraction and was not conditioned on the direction of the reported result.

S2.2 Extraction fields

Sample characteristics. Number of observations, spatial extent, temporal extent, sampling design, response type, prevalence where applicable.

Methodological detail. Algorithms compared, software and version, predictor set, spatial treatment, model complexity settings.

Validation. Partitioning scheme, block size or temporal unit where applicable, whether an

independent test set was available, whether the inference target was stated.

Results. Metrics reported, values, direction of the contrast, whether the comparison was within-study.

Context. Taxon, ecosystem type, management application if any.

Table S6. Studies selected for detailed quantitative extraction.

Study	n	Taxon	System	Comparison	Metrics reported	Result as reported	Quality
Becker et al. (2020)	4,713	Cetaceans	Pelagic, California Current	BRT against GAM; k-fold against leave-one-year-out	Area under the ROC curve; rank correlation	0.85 against 0.83; density surface GAM better on rank correlation; 0.11 partitioning difference	High
Luan et al. (2020)	40 to 120	Fish	Demersal	Random Forest across sample sizes and sampling designs	Performance stability	Distributed samples outperform clustered samples of equal or greater size	High
Mi et al. (2017)	33 to 75	Cranes	Wetland	Random Forest against MaxEnt and BRT	Area under the ROC curve	0.82 to 0.89	High
Thorson et al. (2015)	28 species	Groundfish	Demersal	Geostatistical delta GLMM against design-based estimation	Standard error of index	Approximately 20% reduction	Very high
Thorson (2019a)	About 300 per year	Groundfish	Demersal	Retrospective forecast skill by horizon	Correlation	$r = 0.6$ to 0.7 at one year; 0.3 to 0.4 at five years	High
Hengl et al. (2018)	Multiple benchmarks	Soil	Terrestrial	Spatial Random Forest against kriging	R^2	Approximately 0.41 against approximately 0.60 on Meuse	High
Ploton et al. (2020)	11.8 million	Trees	Terrestrial	Random against spatially structured validation	Conceptual and empirical	Ignoring autocorrelation inflates apparent accuracy	Very high
Mushagalusa et al. (2024)	Simulated	Simulated	Terrestrial	Random against spatial cross-validation	Error magnitude	Spatial partitioning yields larger, more honest errors	High
Grüss et al. (2021)	About 300 per year	Walleye pollock	Demersal, eastern Bering Sea	Spatio-temporal model with environmental covariates	Variance explained; correlation	Cold pool approximately 40% ; cross-validation $r = 0.72$	High
Zelli et al. (2025)	1,247	Corals and sponges	Deep sea, New Zealand	BRT with climate projection	Projected habitat retention	Approximately 23% retained under high emissions	High
Roberts et al. (2017)	Review	Multiple	Multiple	Cross-validation framework	Methodological	Basis for partitioning matched to inference target	Very high

Flynn and Marshall (2013)	Multiple	Lanternfish	Pelagic	GAM with MaxEnt	Application	Deep pelagic application under sparse sampling	Moderate
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S2.3 Quality assessment

Studies were rated across five dimensions: sample adequacy for the inference attempted; validation rigour; reporting clarity, including whether the inference target was stated; statistical robustness; and relevance to data-limited marine settings. Very high denotes exemplary performance on all five; high denotes four or five met with minor limitations; moderate denotes three met adequately. Studies rated low were excluded from quantitative extraction and retained, where relevant, for narrative synthesis only.

Quality ratings determined whether a study could anchor an attributed numerical claim in the main text. They were not used as weights, since no pooling or weighted averaging was performed at any stage.

Appendix S3. Survey design, effort and data quality; environment-specific scaling; two-stage assessment workflows

S3.1 Survey design: coverage, effort and data quality

This appendix gives the extended treatment of the material summarised in main text Section 3.5.

Coverage against total sample size. Spatially dispersed observations spanning the full environmental gradient outperform concentrated samples of equal or larger size (Robinson et al., 2011; Luan et al., 2020). The mechanism is representation of environmental space: clustered samples leave regions unobserved, so prediction there is extrapolation rather than interpolation, and the model is being asked a question its training data cannot answer regardless of how many records that training data contains.

Stratification. Environmental stratification allocates effort proportionally across the gradients structuring the response, typically sea surface temperature, depth zone and productivity regime. Geographic stratification prioritises breadth across regions and is the practical approximation where the structuring gradients are not known in advance. Temporal stratification samples across seasons and years, capturing phenological variation and supplying the replication that spatio-temporal methods require structurally. Under budget constraint, breadth of coverage should be preferred to intensity of local effort (Canonico et al., 2019).

Effort and detection. Sample quality interacts with quantity. Geostatistical models represent varying effort through the likelihood; tree-based methods treat effort as uniform unless it is supplied as a predictor, so an unmodelled effort gradient is absorbed into the fitted habitat relationship. Imperfect detection reduces effective sample size relative to the nominal count, and the reduction is severe where detection probability is low. Taxonomic certainty and covariate matching bear on effective sample size in the same way.

Priorities under constraint. In the order in which they bind: broad spatial coverage ensuring environmental space is represented; sample size meeting the method-specific minimum; tempo-

ral replication where a spatio-temporal method is intended; and environmental covariates at appropriate resolution and temporal alignment.

Below the threshold. With fewer than roughly 30 observations, practitioners should either accept an exploratory analysis with transparent limitations, or defer formal modelling (Guillera-Arroita et al., 2015). Presence-only methods can provide initial insight framed as hypothesis generating rather than hypothesis testing. Expert derived habitat suitability assessment, built from literature and local knowledge, is a legitimate alternative where data are insufficient. Where modelling proceeds, inference should be restricted to coarse scale pattern with uncertainty stated.

S3.2 Environment-specific scaling of minimum sample size

Main text Table 2 gives minimum data requirements by method with environment-specific scaling merged in. The scaling factors are author estimates informed by patterns in the reviewed literature. They are not empirically derived multipliers from controlled experiments, no study in the corpus reports them, and they should be read as an ordering of environments by expected data requirement rather than as coefficients.

Table S7. Environment-specific scaling of minimum sample size requirements. Multipliers are author estimates, not empirical coefficients (grade E).

Environment	Scaling factor	Random Forest minimum	Geostatistical minimum	Rationale
Pelagic	About 1.5 times	About 75 observations	About 45 stations	High spatio-temporal variability from currents, mesoscale eddies and frontal dynamics; mobile species with large ranges; weaker habitat associations
Demersal and benthic	1.0 (reference)	About 50 observations	About 30 stations	Stronger habitat structure and more predictable species substrate associations
Coastal	About 0.75 times	About 40 observations	About 25 stations	Accessible sampling, strong environmental gradients, greater data availability
Deep sea	0.6 to 1.0 times	30 to 50 observations	Not recommended	Extreme data limitation; prefer presence-absence over abundance, and methods that function at minimal sample sizes

The ordering rests on the variability of the environment rather than on accessibility alone. Pelagic systems weaken habitat associations and inflate residual variability (Bell et al., 2014; Robinson et al., 2011); demersal and benthic systems supply the reference case; coastal work benefits from accessibility and steep gradients (Canónico et al., 2019). Deep sea applications face the severest limitation, and the appropriate response is to narrow the inferential target, predicting occurrence

rather than abundance and coarse pattern rather than fine surfaces, rather than to relax the performance standard a model is held to.

S3.3 Index standardisation and two-stage assessment workflows

One class of application warrants separate treatment, because the appropriate method choice is determined by a downstream consumer rather than by the prediction task itself. In two-stage assessment workflows a model is fitted to survey data to produce a time series of annual abundance indices, and those indices are then passed to a separate assessment model that combines them with catch data, biological information and population dynamics assumptions (Thorson, 2019b). The structural assumptions of the downstream consumer constrain the upstream choice.

The binding constraint is the temporal error structure of the indices. Assessment models generally treat the index as a series of observations with independent year-specific errors, and their likelihood, goodness of fit diagnostics, retrospective analysis and uncertainty propagation all rest on that independence. A model that imposes strong temporal autoregressive structure on the abundance field produces indices that are correlated across years, which are not plausible draws from independent year-specific errors. The assessment then understates true uncertainty, because correlated index errors appear more informative than they are, and correlation propagates into estimated trends as retrospective pattern.

This is the reason for a specific structural choice in the geostatistical packages in main text Table 6 (Thorson, 2019b). They permit rich spatial structure through Gaussian Markov random fields on the spatial random effects, which is what makes them useful, while not imposing strong temporal autoregression on the year to year evolution of the abundance field. The annual indices that result are spatially informed and temporally independent in their error structure. Methods that impose explicit AR(1) or random-walk structure on the abundance field yield smoother indices that look more attractive in isolation and are less appropriate for this workflow.

Two qualifications. The recommendation applies to two-stage workflows only. Where the model is itself the final product, in habitat mapping, protected area design, dynamic management or short term forecasting, temporal autoregressive structure may improve prediction at the cost of an independence no downstream consumer requires. And the recommendation concerns temporal structure, not spatial: spatial structure should be included throughout. Practitioners should establish whether model output will feed an assessment before committing to a method's structural choices.

Appendix S4. Computational cost and software

S4.1 Computational cost

Computational cost bears on method selection wherever prediction must be updated frequently, as in dynamic management (main text Table 7). Table S8 gives approximate requirements.

Scaling behaviour matters more than absolute timings, since the latter depend on hardware.

Table S8. Approximate computational requirements. Timings assume typical marine datasets of 100 to 10,000 observations with 5 to 15 predictors, and depend on model complexity, spatial resolution and hardware.

Method	Fitting time	Prediction time per surface	Scaling behaviour
GLM	Seconds	Seconds	Linear
GAM (mgcv)	Seconds to minutes	Seconds	Sub linear with bam
Random Forest (ranger)	Seconds to minutes	Seconds	Linear in observations and predictors
BRT (gbm)	Minutes	Seconds	Linear in tree number
VAST	Minutes to hours	Minutes	Super linear in knots
sdmTMB	Minutes to about 30 minutes	Seconds to minutes	Super linear in mesh complexity
tinyVAST	Minutes to hours	Seconds to minutes	Super linear in mesh complexity and species number
BART	Minutes	Seconds	Linear in tree number
Hmsc	Hours; MCMC with 2 to 4 chains and 1,000 or more posterior samples	Minutes for posterior predictive draws across the prediction grid	Super linear in species richness and site count
sjSDM	Minutes to about 30 minutes; variational inference	Seconds to minutes	Linear to sub linear in species count; GPU accelerable

The operational consequence is the one drawn in main text Table 7. Tree-based methods dominate near real-time systems because prediction is fast and requires no mesh, not because they predict better. Where calibrated uncertainty is required in such a system, the cost of a geostatistical alternative falls on fitting rather than prediction, which is why separating frequent field updates from infrequent hyperparameter re estimation is the plausible route to operational deployment.

S4.2 Software and implementation

Table S9. Software implementations for the method families in main text Table 6, with the functions that carry the spatial component and notes specific to marine application. Package versions change; access dates are September 2026.

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Random Forest	ranger; randomForest	ranger() for fast fitting with variable importance; randomForest() as the reference implementation; spatial predictors are constructed by the analyst	scikit-learn	ranger preferred above roughly 10,000 observations for its C++ backend; use permutation importance where predictors are correlated, which is usual with marine covariates
Spatial Random Forest	spatialRF; RFsp via landmap; waywiser; RandomForestsGLS	rf_spatial() fits with spatial predictors; rf_evaluate() for spatial evaluation; RandomForestsGLS provides the dependent data formulation	verstack	Distance predictors must be computed against an ecologically meaningful reference, not the coordinate grid alone; distance to shore, shelf break or depth contour are the usual choices
VAST	VAST; FishStatsUtils; TMB (Kristensen et al., 2016)	fit_model() is the primary interface; make_data() and make_model() for configuration	limited	Knot number controls both the resolution of the spatial field and the fitting cost; start coarse and refine, since super linear scaling makes over specification expensive
sdmTMB	sdmTMB (Anderson et al., 2025); TMB; fmesher	sdmTMB() uses a formula interface comparable to glm or mgcv; make_mesh() constructs the SPDE mesh; sanity() reports convergence checks	limited	The recommended entry point for practitioners moving from mgcv; delta families handle the zero inflation typical of trawl and transect data; get_index() produces standardised abundance indices
tinyVAST	tinyVAST (Thorson et al., 2025); TMB; fmesher	tinyVAST() extends the sdmTMB conventions to multivariate and structural equation specifications	limited	For multi-species work without VAST's full specification overhead; cross species spatial fields borrow strength toward sparsely observed species
Spatial GAMs	mgcv (Wood, 2017); dsm (Miller et al., 2022); gratia	gam() and bam() with s(x, y, bs = "tp") or te(x, y); dsm for density surface models	pyGAM	dsm is the standard route for line transect survey data, which covers most cetacean and seabird applications
MaxEnt	maxnet; ENMeval; dismo	maxnet() is the native R implementation requiring no Java; ENMevaluate() automates tuning and background selection	elapid	Background extent defines the domain against which discrimination is measured and must match the area over which prediction is required; it drives apparent performance more than any tuning setting
Boosted Regression Trees	gbm; dismo; xgboost; lightgbm; caret	gbm.step() performs cross-validated selection of tree number with ecologically appropriate defaults (Elith et al., 2008)	xgboost, lightgbm	gbm.step() applies ecologically appropriate defaults and cross-validated tree selection; prefer it to direct fitting for distribution work

Table S9 (continued)

Method	R packages	Key functions and workflow	Python	Notes for marine applications
BART	embarcadero; dbarts	bart.step() for variable selection in a distribution modelling context; bart() and bart2() for fitting	bartpy	Posterior uncertainty without MCMC cost at community scale
Graphical interface	GLOSSA (Mestre-Tomás et al., 2025)	Shiny application providing a no code interface to BART based models with spatial blocking	none	Useful where analysis must be handed to non specialist staff, or where a workflow is demonstrated to stakeholders
Joint SDM, hierarchical	Hmsc	Hmsc() with spatial random effects; createStudyDesign() takes spatial coordinates; computeAssociations() returns residual species correlations; MCMC sampling, hours for large communities	limited	Benefit concentrates where a rare target species co occurs with better sampled congeners; see Table S10
Joint SDM, variational	sjSDM (Pichler and Hartig, 2021)	sjSDM() performs fast variational inference through a PyTorch backend	via the same backend	Faster than full MCMC for large communities, at some cost in posterior accuracy; see Table S10
Joint SDM, ordination	boral	boral() for Bayesian latent variable ordination of multivariate abundance data	limited	Ordination and community description rather than spatial prediction; see Table S10
Ensembles	biomod2; caretEnsemble; stacks	BIOMOD_Modeling() and BIOMOD_EnsembleModeling() for multi model platforms	scikit-learn	Appropriate where candidates are inseparable under matched validation (main text 3.4), not as a default
Spatial cross-validation	blockCV (Valavi et al., 2019); spatialsample; ENMeval	cv_spatial() performs block partitioning with automated range estimation; cv_spatial_autocor() estimates the range; cv_cluster() supplies spatial clustering where blocks cannot be formed; block size should exceed the residual autocorrelation range	verstack, spacy	Block size must exceed the residual autocorrelation range, which should be estimated rather than assumed; where the subsample extent is smaller than the range, blocking at the recommended size is impossible and this should be reported
Spatial diagnostics	spdep; gstat; DHARMA	moran.test() and moran.mc() for Moran's I on residuals; variogram() for empirical variograms; residual simulation for distributional checks	pysal	Moran's I on residuals rather than on the raw response; residual structure is what bears on the model

Table S9 (continued)

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Graph neural networks	torch	Limited R support; most implementations use Python	PyTorch Geometric	Noted for completeness; data requirements exceed the settings this review addresses

Table S10. Joint species distribution model frameworks, distinguished by where each applies shrinkage and how inference is performed. The distinction matters under data limitation, because shrinkage on environmental responses is the mechanism by which sparsely observed species borrow strength from better sampled ones.

Framework	R package	Shrinkage on environmental responses	Shrinkage on spatial latent variables	Inference	Recommended marine use
Hierarchical, with spatial random effects	Hmsc	Yes; improves estimation for species with sparse observations	Yes; captures residual spatial autocorrelation across the community	Full MCMC; hours	Demersal fish and benthic assemblages; community level conservation planning where a full posterior is required. Norberg et al. (2019) report the spatial variant outperforming non-spatial joint models
Hierarchical, without spatial random effects	Hmsc	Yes	No	Full MCMC	Only where residual spatial autocorrelation is genuinely weak, or where computation precludes spatial latent factors; otherwise prefer the spatial variant
Variational	sjSDM (Pichler and Hartig, 2021)	Yes, through regularisation	Yes, through spatial latent variables	Variational inference; minutes	Large communities above roughly 30 species; iterative assessment against tight timelines; faster than full MCMC at some cost in posterior accuracy
Bayesian ordination	boral	Yes, through latent variables	No spatial structure in the base implementation	Full MCMC	Compositional analysis and ordination; community description rather than spatial prediction
Multi-species geostatistical	VAST (Thorson, 2019b)	Implicit, through shared random effects	Yes; vector autoregressive structure on spatial fields across species	Empirical Bayes by Laplace approximation; minutes to hours	Multi-species survey data from structured programmes, and where output feeds an assessment model (Appendix S3.3)
Multi-species geostatistical, formula interface	tinyVAST (Thorson et al., 2025)	Implicit, through shared random effects	Yes; cross species spatial fields with optional simultaneous effects	Empirical Bayes; minutes to hours	For practitioners moving from sdmTMB to multi-species work without the full specification overhead

Appendix S5. Reporting protocol

We recommend that primary studies contributing to future syntheses report, at minimum: the inference target the validation was designed to estimate; the partitioning scheme and, for spatial blocking, the block size and its relation to the estimated autocorrelation range, including whether the study extent permitted blocking at that size; at least one discrimination metric and one error or calibration metric where the response type permits both; the spatial and environmental coverage of training data relative to the prediction domain; and the proportion of the prediction domain requiring extrapolation. The ODMAP protocol (Zurell et al., 2020) provides the structure for the first four; the metric convention is the missing element.

Appendix S6. Benchmark sampling and validation designs

All panels were drawn by `figure_sampling_designs.R`, which sources the benchmark scaffold's partition, subsampling and fold functions and applies them to the survey hauls with a figure-specific seed (20260925), so what is mapped is what the benchmark draws. The species shown is Dover sole. Extent, printed in each subsample panel, is the area of the subsample's bounding box as a fraction of the training pool's bounding box and is the quantity the benchmark records as `extent_frac`; Table S17 lists it, together with the number of redraws and the number of presences, for every panel. Colour meanings are the same in every design figure: grey for pool hauls not drawn, black for the training subsample, and a single reddish purple for the evaluation set; none of these is a model colour. Land is shaded and a 200 km scale bar, exact at its latitude, is drawn in the first panel of each figure.

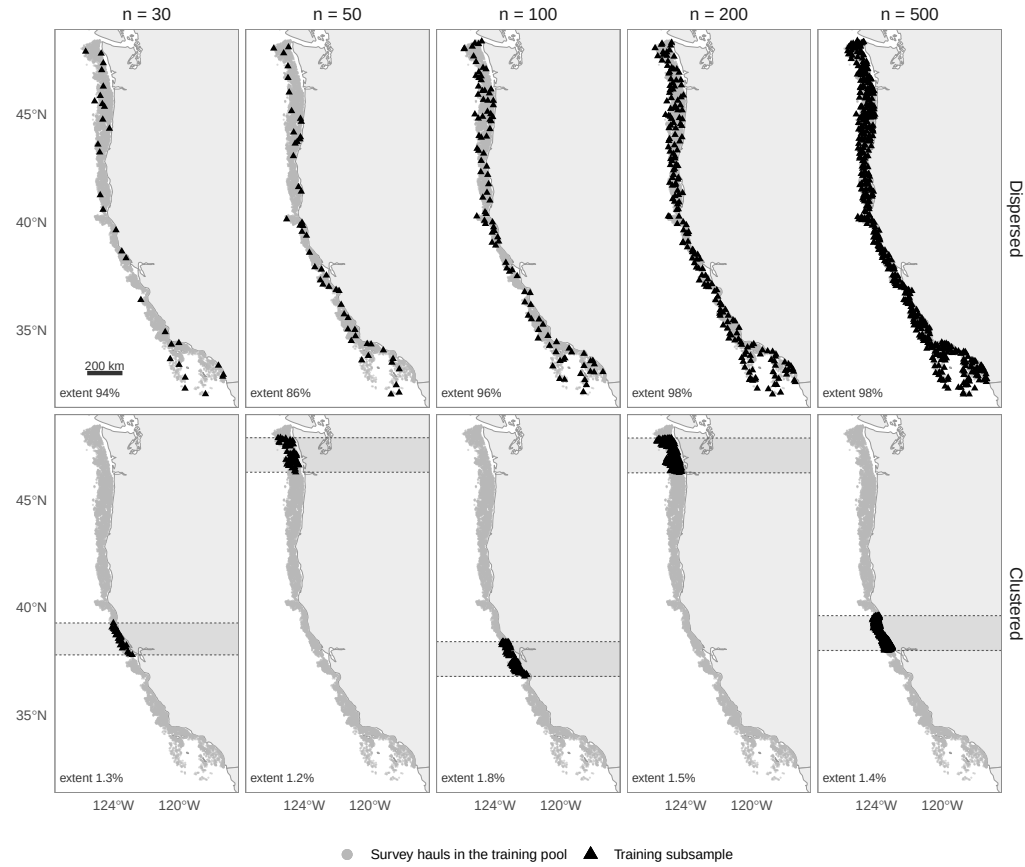


Figure S1. Example training subsamples for every cell of the conditional target design. Rows are the coverage designs, columns the five sample sizes. The conditional evaluation set (a random 30 percent of hauls interleaved with the pool) is omitted for legibility and is shown in main text Figure 2. Band limits of clustered draws are dashed and shaded. Dispersed draws span 85 to 99 percent of the pool's bounding box at every sample size and clustered draws 1 to 2 percent, so coverage is controlled independently of sample size as the design intends.

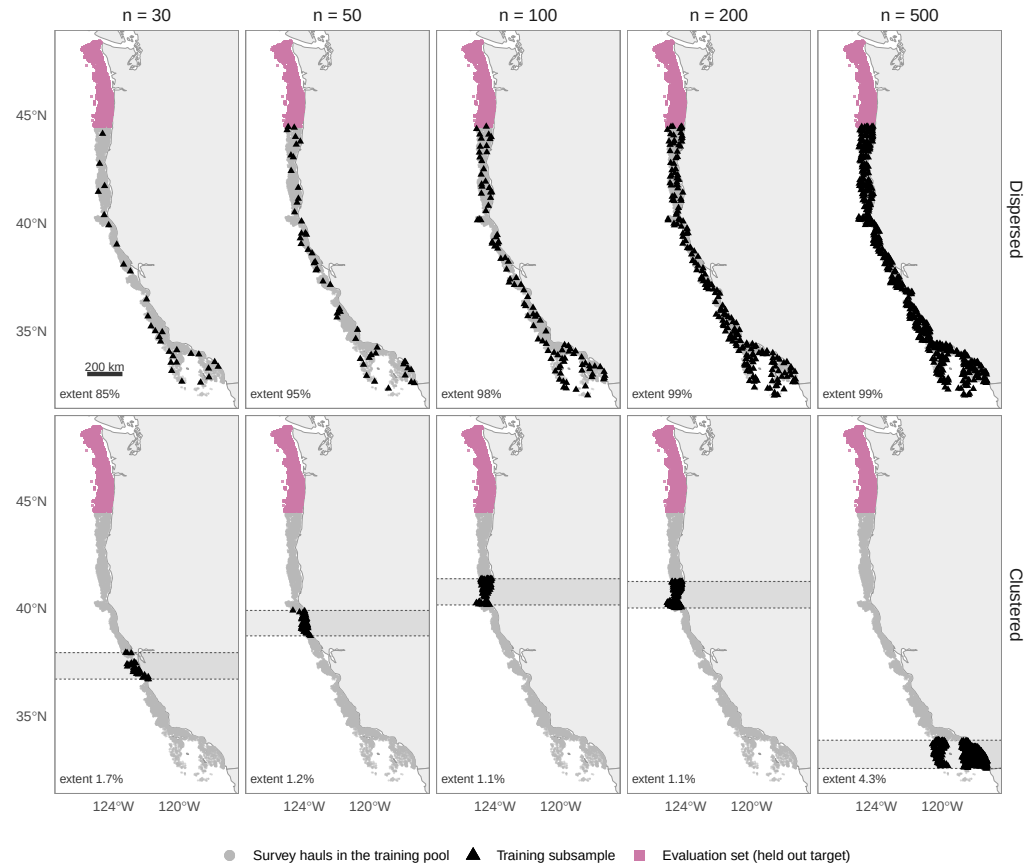


Figure S2. Example training subsamples for every cell of the marginal target design. As Figure S1, with the northern evaluation region shown so that the extrapolation distance from each training subsample to the target is visible. The clustered band at $n = 500$ is the exception to the 1 to 2 percent extent, at 4 percent, because the survey widens across the Southern California Bight.

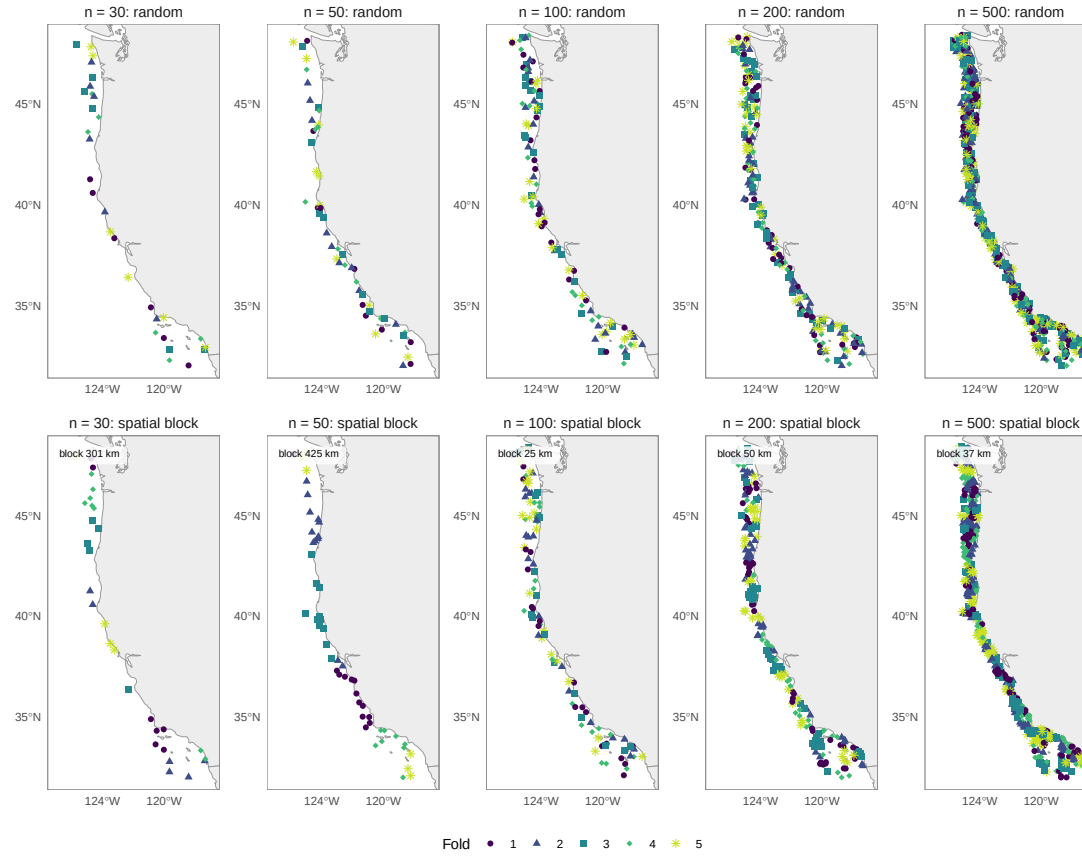


Figure S3. Random and spatial block folds for dispersed training subsamples at every sample size (conditional target pool). Upper row random five-fold, lower row spatial block five-fold with the block side printed in each panel. The block side is the autocorrelation range of the residuals of a covariate-only GAM fitted to the subsample, capped at the subsample extent divided by the ceiling of the square root of the number of folds. It varied from 25 to 425 km across sample sizes with no trend, because the range is poorly determined at small n . Where it is large the folds are contiguous stretches of coast; where it is small they approach a random partition. Fold colours and shapes identify folds only.

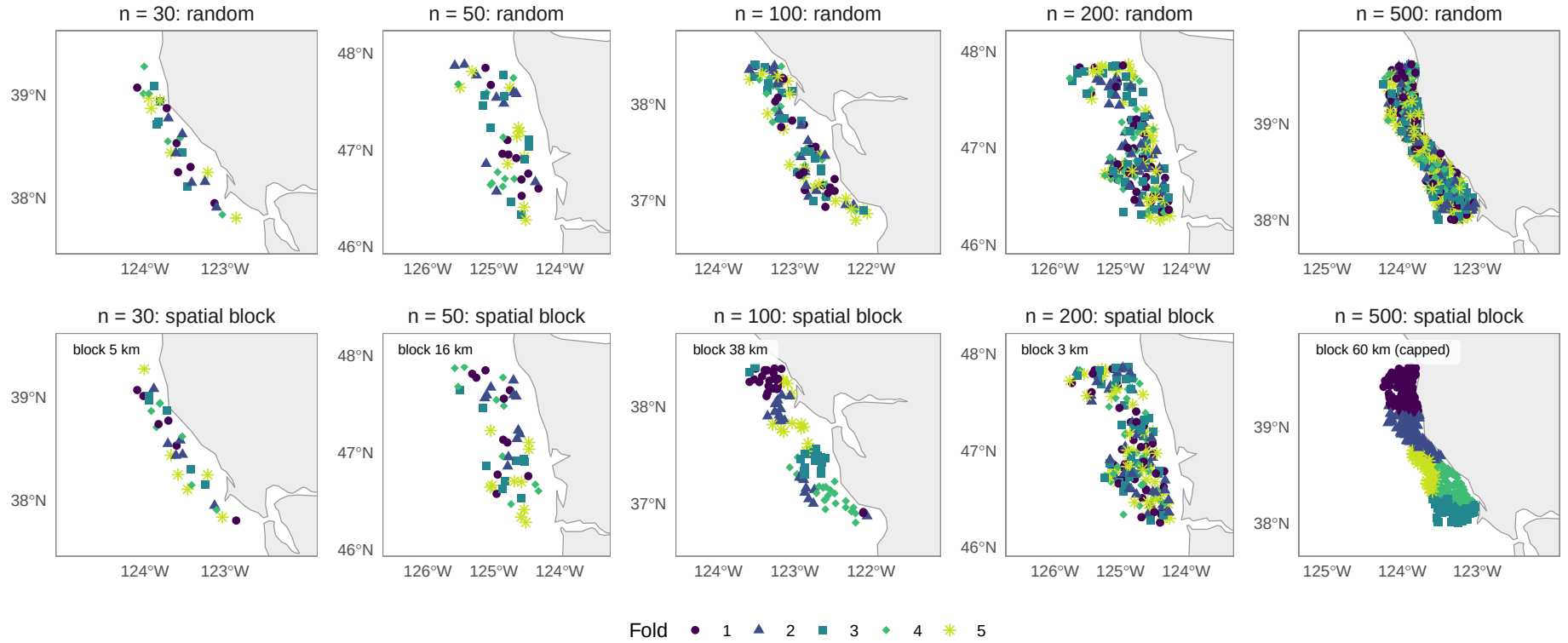


Figure S4. Random and spatial block folds for clustered training subsamples at every sample size (conditional target pool). Each panel is zoomed to its own band. Estimated block sides were 3 to 60 km; only the largest, at $n = 500$, produced folds that separate the band into distinct sections, and that one was capped by the extent rule. At every other sample size the spatial partition is indistinguishable from the random one, which is why the two validation estimates agree so closely under clustered coverage in main text Section 3.4.3.

Table S17. Draws used for the design figures: target, coverage, sample size, extent, redraws and presences for every mapped panel.

Target	Coverage	n	Species	Extent	Redraws	Presences
conditional	dispersed	30	dover sole	0.936	1	21
marginal	dispersed	30	dover sole	0.851	1	25
conditional	clustered	30	dover sole	0.013	0	24
marginal	clustered	30	dover sole	0.017	0	25
conditional	dispersed	50	dover sole	0.858	0	38
marginal	dispersed	50	dover sole	0.947	0	38
conditional	clustered	50	dover sole	0.012	2	45
marginal	clustered	50	dover sole	0.012	1	38
conditional	dispersed	100	dover sole	0.960	0	71
marginal	dispersed	100	dover sole	0.985	0	80
conditional	clustered	100	dover sole	0.018	0	83
marginal	clustered	100	dover sole	0.011	0	87
conditional	dispersed	200	dover sole	0.976	0	153
marginal	dispersed	200	dover sole	0.988	0	150
conditional	clustered	200	dover sole	0.015	0	174
marginal	clustered	200	dover sole	0.011	0	181
conditional	dispersed	500	dover sole	0.977	0	397
marginal	dispersed	500	dover sole	0.991	0	407
conditional	clustered	500	dover sole	0.014	0	441
marginal	clustered	500	dover sole	0.043	0	346

Appendix S7. Benchmark additional results

Performance summaries exclude replicates with an error, no finite prediction or a degenerate prediction surface (standard deviation of predictions below 0.001 across the evaluation set). Intervals are t-based over the pooled replicates of the three species. Paired contrasts are within-replicate differences between two models fitted to the same subsample and evaluated on the same held-out set (Tables S13 and S17a), or between the two coverage designs at the same species, replicate, sample size and target (Table S13). The mixed model check (Table S16a) was fitted with lme4 with random intercepts for species and for replicate within species; with three species the species variance component is weakly identified (standard deviation 0.04 against a residual standard deviation of 0.10) and the model’s constant residual variance makes its intervals conservative, for the reason given in main text Section 2.5. All values are from the run of 25 September 2026 (10 replicates per cell) and regenerate from results_all.rds through route2_analysis.R.

Table S11. Mean held-out AUC by species, model and sample size under dispersed coverage and the conditional target (usable fits only).

Species	Model	n = 30	n = 50	n = 100	n = 200	n = 500
canary rockfish	RF	0.798	0.806	0.787	0.793	0.849
canary rockfish	RF + coordinates	0.809	0.811	0.819	0.817	0.871
canary rockfish	GAM	0.767	0.781	0.814	0.842	0.880
canary rockfish	GAM + s(X,Y)	0.648	0.679	0.658	0.825	0.886
canary rockfish	GMRF	0.797	0.820	0.799	0.812	0.872
dover sole	RF	0.779	0.769	0.807	0.827	0.839
dover sole	RF + coordinates	0.820	0.819	0.852	0.869	0.893
dover sole	GAM	0.730	0.768	0.819	0.843	0.859
dover sole	GAM + s(X,Y)	0.697	0.680	0.813	0.853	0.877
dover sole	GMRF	0.757	0.796	0.826	0.850	0.866
lingcod	RF	0.843	0.852	0.859	0.867	0.876
lingcod	RF + coordinates	0.843	0.862	0.877	0.885	0.893
lingcod	GAM	0.813	0.841	0.857	0.867	0.874
lingcod	GAM + s(X,Y)	0.702	0.750	0.839	0.886	0.892
lingcod	GMRF	0.813	0.862	0.869	0.874	0.889



Figure S5. Held-out AUC by model at each sample size in the benchmark: mean and 95 percent interval over replicates, dodged within each sample size; panels are coverage design by inference target.

Table S12. Paired contrasts in held-out AUC for every cell of the benchmark design: mean, 95 percent interval, number of replicates and share of replicates with a positive difference, for the five contrasts of main text Section 2.5.

Contrast	n	Coverage	Target	Replicates	Mean	95% lower	95% upper	Share positive
rf_spatial - rf (spatial effect, RF)	30	dispersed	conditional	29	0.017	0.003	0.031	0.66
rf_spatial - rf (spatial effect, RF)	30	dispersed	marginal	24	0.023	-0.001	0.046	0.50
rf_spatial - rf (spatial effect, RF)	30	clustered	conditional	30	-0.040	-0.068	-0.012	0.43
rf_spatial - rf (spatial effect, RF)	30	clustered	marginal	30	0.000	-0.023	0.023	0.33
rf_spatial - rf (spatial effect, RF)	50	dispersed	conditional	30	0.022	0.007	0.036	0.80
rf_spatial - rf (spatial effect, RF)	50	dispersed	marginal	30	0.004	-0.021	0.028	0.50
rf_spatial - rf (spatial effect, RF)	50	clustered	conditional	30	-0.020	-0.045	0.005	0.53
rf_spatial - rf (spatial effect, RF)	50	clustered	marginal	30	0.005	-0.018	0.028	0.23
rf_spatial - rf (spatial effect, RF)	100	dispersed	conditional	30	0.031	0.025	0.038	0.97
rf_spatial - rf (spatial effect, RF)	100	dispersed	marginal	30	0.017	0.003	0.031	0.70
rf_spatial - rf (spatial effect, RF)	100	clustered	conditional	30	-0.026	-0.059	0.007	0.50
rf_spatial - rf (spatial effect, RF)	100	clustered	marginal	30	0.004	-0.014	0.023	0.47
rf_spatial - rf (spatial effect, RF)	200	dispersed	conditional	30	0.028	0.022	0.035	0.97
rf_spatial - rf (spatial effect, RF)	200	dispersed	marginal	30	0.013	0.001	0.025	0.73
rf_spatial - rf (spatial effect, RF)	200	clustered	conditional	30	-0.019	-0.041	0.004	0.40
rf_spatial - rf (spatial effect, RF)	200	clustered	marginal	30	-0.009	-0.041	0.023	0.43
rf_spatial - rf (spatial effect, RF)	500	dispersed	conditional	30	0.031	0.024	0.038	1.00
rf_spatial - rf (spatial effect, RF)	500	dispersed	marginal	30	0.012	-0.003	0.026	0.67
rf_spatial - rf (spatial effect, RF)	500	clustered	conditional	30	0.001	-0.019	0.021	0.57
rf_spatial - rf (spatial effect, RF)	500	clustered	marginal	30	0.001	-0.017	0.018	0.50
gam_spatial - gam (spatial effect, GAM)	30	dispersed	conditional	29	-0.087	-0.128	-0.045	0.17
gam_spatial - gam (spatial effect, GAM)	30	dispersed	marginal	22	-0.085	-0.155	-0.015	0.27
gam_spatial - gam (spatial effect, GAM)	30	clustered	conditional	30	-0.135	-0.191	-0.079	0.27
gam_spatial - gam (spatial effect, GAM)	30	clustered	marginal	12	-0.133	-0.202	-0.063	0.17
gam_spatial - gam (spatial effect, GAM)	50	dispersed	conditional	30	-0.094	-0.130	-0.058	0.13

Table S12 (continued)

Contrast	n	Coverage	Target	Replicates	Mean	95% lower	95% upper	Share positive
gam_spatial - gam (spatial effect, GAM)	50	dispersed	marginal	27	-0.080	-0.127	-0.033	0.22
gam_spatial - gam (spatial effect, GAM)	50	clustered	conditional	30	-0.151	-0.196	-0.106	0.17
gam_spatial - gam (spatial effect, GAM)	50	clustered	marginal	12	-0.095	-0.182	-0.009	0.17
gam_spatial - gam (spatial effect, GAM)	100	dispersed	conditional	30	-0.060	-0.094	-0.026	0.40
gam_spatial - gam (spatial effect, GAM)	100	dispersed	marginal	29	-0.105	-0.148	-0.063	0.21
gam_spatial - gam (spatial effect, GAM)	100	clustered	conditional	30	-0.138	-0.185	-0.091	0.13
gam_spatial - gam (spatial effect, GAM)	100	clustered	marginal	20	-0.064	-0.118	-0.010	0.35
gam_spatial - gam (spatial effect, GAM)	200	dispersed	conditional	30	0.004	-0.015	0.022	0.83
gam_spatial - gam (spatial effect, GAM)	200	dispersed	marginal	29	-0.042	-0.067	-0.016	0.38
gam_spatial - gam (spatial effect, GAM)	200	clustered	conditional	30	-0.102	-0.142	-0.062	0.10
gam_spatial - gam (spatial effect, GAM)	200	clustered	marginal	20	-0.031	-0.065	0.004	0.45
gam_spatial - gam (spatial effect, GAM)	500	dispersed	conditional	30	0.014	0.010	0.018	0.90
gam_spatial - gam (spatial effect, GAM)	500	dispersed	marginal	30	-0.013	-0.027	0.001	0.53
gam_spatial - gam (spatial effect, GAM)	500	clustered	conditional	30	-0.101	-0.147	-0.055	0.17
gam_spatial - gam (spatial effect, GAM)	500	clustered	marginal	18	-0.008	-0.026	0.011	0.44
gmrf - gam_spatial (GMRF vs smooth surface)	30	dispersed	conditional	19	0.093	0.052	0.134	0.89
gmrf - gam_spatial (GMRF vs smooth surface)	30	dispersed	marginal	16	0.124	0.051	0.198	0.88
gmrf - gam_spatial (GMRF vs smooth surface)	30	clustered	conditional	29	0.145	0.091	0.198	0.79
gmrf - gam_spatial (GMRF vs smooth surface)	30	clustered	marginal	9	0.123	0.038	0.208	1.00
gmrf - gam_spatial (GMRF vs smooth surface)	50	dispersed	conditional	13	0.082	0.029	0.135	0.69
gmrf - gam_spatial (GMRF vs smooth surface)	50	dispersed	marginal	20	0.069	0.022	0.115	0.75

Table S12 (continued)

Contrast	n	Coverage	Target	Replicates	Mean	95% lower	95% upper	Share positive
gmrf - gam_spatial (GMRF vs smooth surface)	50	clustered	conditional	30	0.138	0.089	0.188	0.87
gmrf - gam_spatial (GMRF vs smooth surface)	50	clustered	marginal	11	0.093	-0.001	0.186	0.82
gmrf - gam_spatial (GMRF vs smooth surface)	100	dispersed	conditional	23	0.055	0.020	0.089	0.70
gmrf - gam_spatial (GMRF vs smooth surface)	100	dispersed	marginal	20	0.076	0.033	0.120	0.80
gmrf - gam_spatial (GMRF vs smooth surface)	100	clustered	conditional	26	0.114	0.065	0.162	0.85
gmrf - gam_spatial (GMRF vs smooth surface)	100	clustered	marginal	20	0.007	-0.066	0.081	0.55
gmrf - gam_spatial (GMRF vs smooth surface)	200	dispersed	conditional	25	-0.007	-0.028	0.015	0.24
gmrf - gam_spatial (GMRF vs smooth surface)	200	dispersed	marginal	26	0.004	-0.025	0.033	0.46
gmrf - gam_spatial (GMRF vs smooth surface)	200	clustered	conditional	30	0.096	0.050	0.142	0.80
gmrf - gam_spatial (GMRF vs smooth surface)	200	clustered	marginal	19	0.030	-0.034	0.094	0.58
gmrf - gam_spatial (GMRF vs smooth surface)	500	dispersed	conditional	30	-0.009	-0.013	-0.006	0.07
gmrf - gam_spatial (GMRF vs smooth surface)	500	dispersed	marginal	29	-0.024	-0.051	0.003	0.38
gmrf - gam_spatial (GMRF vs smooth surface)	500	clustered	conditional	30	0.098	0.053	0.143	0.80
gmrf - gam_spatial (GMRF vs smooth surface)	500	clustered	marginal	18	-0.000	-0.022	0.022	0.56

Table S12 (continued)

Contrast	n	Coverage	Target	Replicates	Mean	95% lower	95% upper	Share positive
rf - gam (algorithm, non-spatial)	30	dispersed	conditional	29	0.037	0.002	0.072	0.59
rf - gam (algorithm, non-spatial)	30	dispersed	marginal	24	0.045	0.009	0.082	0.54
rf - gam (algorithm, non-spatial)	30	clustered	conditional	30	0.032	0.006	0.058	0.70
rf - gam (algorithm, non-spatial)	30	clustered	marginal	30	0.048	0.012	0.085	0.60
rf - gam (algorithm, non-spatial)	50	dispersed	conditional	30	0.012	-0.010	0.034	0.43
rf - gam (algorithm, non-spatial)	50	dispersed	marginal	30	0.031	0.005	0.057	0.63
rf - gam (algorithm, non-spatial)	50	clustered	conditional	30	0.018	-0.009	0.044	0.57
rf - gam (algorithm, non-spatial)	50	clustered	marginal	30	0.050	0.013	0.086	0.63
rf - gam (algorithm, non-spatial)	100	dispersed	conditional	30	-0.012	-0.029	0.004	0.23
rf - gam (algorithm, non-spatial)	100	dispersed	marginal	30	-0.011	-0.029	0.008	0.43
rf - gam (algorithm, non-spatial)	100	clustered	conditional	30	0.004	-0.028	0.035	0.47
rf - gam (algorithm, non-spatial)	100	clustered	marginal	30	0.024	-0.011	0.058	0.50
rf - gam (algorithm, non-spatial)	200	dispersed	conditional	30	-0.022	-0.034	-0.010	0.20
rf - gam (algorithm, non-spatial)	200	dispersed	marginal	30	-0.012	-0.029	0.005	0.43
rf - gam (algorithm, non-spatial)	200	clustered	conditional	30	-0.021	-0.045	0.004	0.43
rf - gam (algorithm, non-spatial)	200	clustered	marginal	30	0.020	-0.016	0.056	0.50
rf - gam (algorithm, non-spatial)	500	dispersed	conditional	30	-0.017	-0.022	-0.011	0.17
rf - gam (algorithm, non-spatial)	500	dispersed	marginal	30	-0.007	-0.017	0.003	0.30
rf - gam (algorithm, non-spatial)	500	clustered	conditional	30	-0.031	-0.051	-0.011	0.13
rf - gam (algorithm, non-spatial)	500	clustered	marginal	30	0.019	-0.017	0.055	0.27
rf_spatial - gam_spatial (algorithm, spatial)	30	dispersed	conditional	29	0.141	0.112	0.169	1.00
rf_spatial - gam_spatial (algorithm, spatial)	30	dispersed	marginal	22	0.157	0.112	0.203	0.95
rf_spatial - gam_spatial (algorithm, spatial)	30	clustered	conditional	30	0.127	0.071	0.183	0.83
rf_spatial - gam_spatial (algorithm, spatial)	30	clustered	marginal	12	0.153	0.067	0.238	0.92
rf_spatial - gam_spatial (algorithm, spatial)	50	dispersed	conditional	30	0.128	0.095	0.160	0.90
rf_spatial - gam_spatial (algorithm, spatial)	50	dispersed	marginal	27	0.108	0.067	0.149	0.93

Table S12 (continued)

Contrast	n	Coverage	Target	Replicates	Mean	95% lower	95% upper	Share positive
rf_spatial - gam_spatial (algorithm, spatial)	50	clustered	conditional	30	0.149	0.100	0.197	0.83
rf_spatial - gam_spatial (algorithm, spatial)	50	clustered	marginal	12	0.161	0.086	0.236	1.00
rf_spatial - gam_spatial (algorithm, spatial)	100	dispersed	conditional	30	0.079	0.049	0.109	0.87
rf_spatial - gam_spatial (algorithm, spatial)	100	dispersed	marginal	29	0.111	0.070	0.151	0.72
rf_spatial - gam_spatial (algorithm, spatial)	100	clustered	conditional	30	0.115	0.063	0.168	0.80
rf_spatial - gam_spatial (algorithm, spatial)	100	clustered	marginal	20	0.088	0.031	0.146	0.75
rf_spatial - gam_spatial (algorithm, spatial)	200	dispersed	conditional	30	0.003	-0.015	0.020	0.60
rf_spatial - gam_spatial (algorithm, spatial)	200	dispersed	marginal	29	0.043	0.012	0.073	0.66
rf_spatial - gam_spatial (algorithm, spatial)	200	clustered	conditional	30	0.062	0.012	0.112	0.67
rf_spatial - gam_spatial (algorithm, spatial)	200	clustered	marginal	20	0.038	-0.029	0.105	0.45
rf_spatial - gam_spatial (algorithm, spatial)	500	dispersed	conditional	30	0.001	-0.005	0.006	0.60
rf_spatial - gam_spatial (algorithm, spatial)	500	dispersed	marginal	30	0.018	-0.006	0.041	0.47
rf_spatial - gam_spatial (algorithm, spatial)	500	clustered	conditional	30	0.071	0.020	0.122	0.60
rf_spatial - gam_spatial (algorithm, spatial)	500	clustered	marginal	18	0.029	-0.038	0.096	0.56

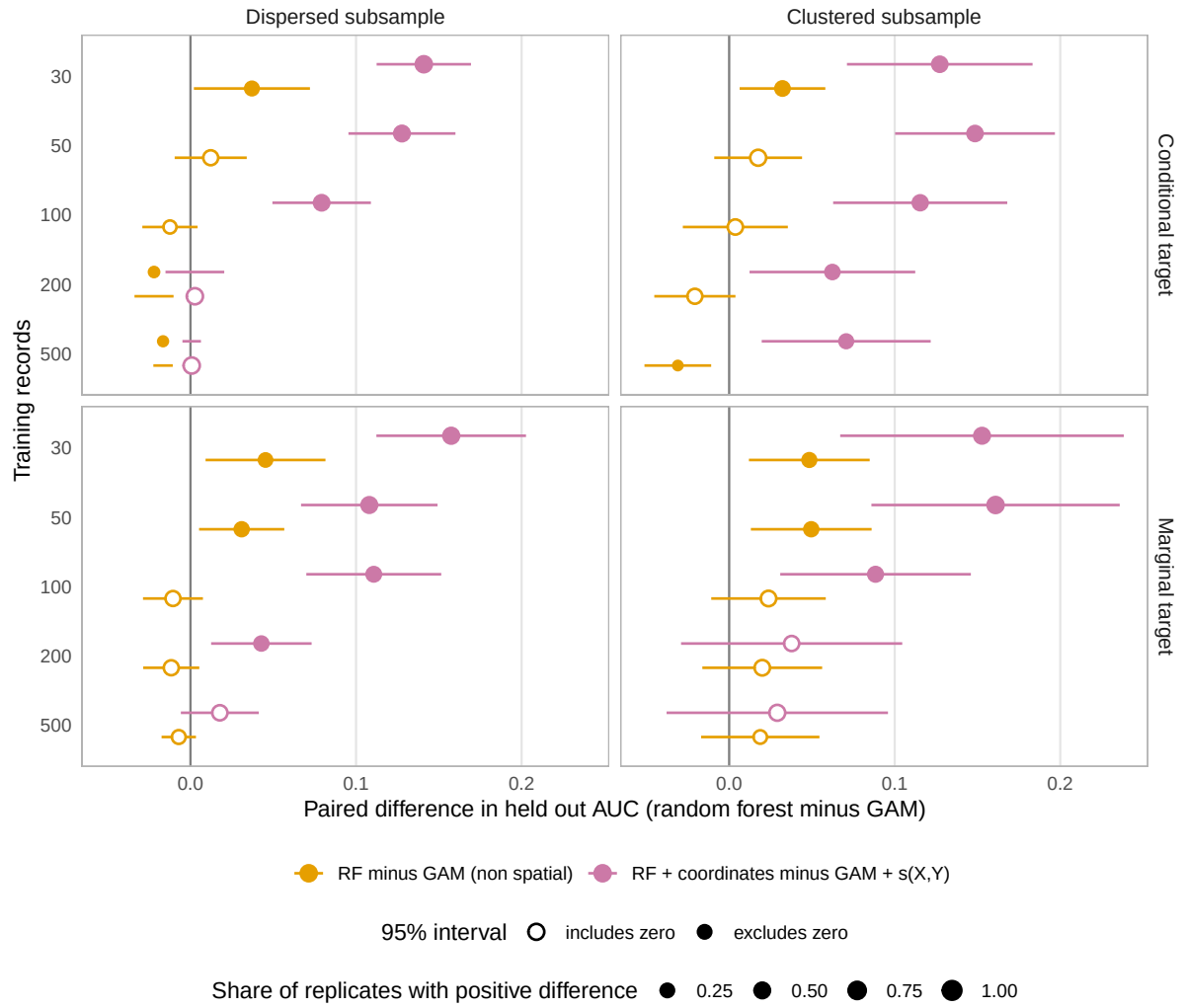


Figure S6. Algorithm contrasts as forest plots: random forest minus GAM without spatial terms, and random forest with coordinates minus GAM with a coordinate smooth. Rows are sample sizes, whiskers the paired 95 percent interval, filled markers where the interval excludes zero, marker size the share of replicates with a positive difference.

Table S13. Coverage effect (dispersed minus clustered) in held-out AUC by model, sample size and target.

Model	n	Target	Replicates	Mean	95% lower	95% upper	Share positive
RF	30	conditional	29	0.045	0.011	0.078	0.76
RF + coordinates	30	conditional	29	0.096	0.053	0.139	0.83
GAM	30	conditional	29	0.041	-0.003	0.086	0.66
GAM + s(X,Y)	30	conditional	29	0.082	0.043	0.121	0.79
GMRF	30	conditional	18	0.044	-0.003	0.092	0.67
RF	30	marginal	24	0.075	-0.007	0.157	0.54
RF + coordinates	30	marginal	24	0.087	0.003	0.171	0.54
GAM	30	marginal	24	0.090	0.006	0.174	0.58
GAM + s(X,Y)	30	marginal	9	-0.025	-0.114	0.065	0.33
GMRF	30	marginal	16	0.093	-0.033	0.218	0.56
RF	50	conditional	30	0.039	0.016	0.062	0.73
RF + coordinates	50	conditional	30	0.081	0.042	0.119	0.77
GAM	50	conditional	30	0.045	0.004	0.085	0.57
GAM + s(X,Y)	50	conditional	30	0.102	0.046	0.157	0.77
GMRF	50	conditional	13	0.081	0.022	0.140	0.85
RF	50	marginal	30	0.045	-0.013	0.103	0.50
RF + coordinates	50	marginal	30	0.044	-0.018	0.105	0.43
GAM	50	marginal	30	0.064	-0.021	0.148	0.63
GAM + s(X,Y)	50	marginal	11	0.130	-0.002	0.262	0.82
GMRF	50	marginal	21	0.066	-0.024	0.156	0.52
RF	100	conditional	30	0.030	0.010	0.050	0.73
RF + coordinates	100	conditional	30	0.088	0.044	0.131	0.97
GAM	100	conditional	30	0.046	0.007	0.085	0.63
GAM + s(X,Y)	100	conditional	30	0.124	0.071	0.177	0.83
GMRF	100	conditional	22	0.075	0.035	0.114	0.86
RF	100	marginal	30	0.052	-0.003	0.107	0.57
RF + coordinates	100	marginal	30	0.064	0.004	0.125	0.57

Table S13 (continued)

Model	n	Target	Replicates	Mean	95% lower	95% upper	Share positive
GAM	100	marginal	30	0.086	0.004	0.169	0.67
GAM + s(X,Y)	100	marginal	20	-0.001	-0.107	0.104	0.55
GMRF	100	marginal	21	0.099	0.003	0.195	0.67
RF	200	conditional	30	0.032	0.013	0.051	0.87
RF + coordinates	200	conditional	30	0.079	0.054	0.105	1.00
GAM	200	conditional	30	0.033	0.008	0.059	0.77
GAM + s(X,Y)	200	conditional	30	0.139	0.083	0.194	0.97
GMRF	200	conditional	25	0.039	0.014	0.065	0.84
RF	200	marginal	30	0.025	-0.025	0.075	0.37
RF + coordinates	200	marginal	30	0.047	-0.015	0.109	0.63
GAM	200	marginal	30	0.057	-0.009	0.123	0.50
GAM + s(X,Y)	200	marginal	19	0.069	-0.038	0.175	0.58
GMRF	200	marginal	26	-0.014	-0.032	0.004	0.38
RF	500	conditional	30	0.062	0.039	0.086	1.00
RF + coordinates	500	conditional	30	0.092	0.067	0.117	1.00
GAM	500	conditional	30	0.048	0.023	0.072	0.93
GAM + s(X,Y)	500	conditional	30	0.162	0.104	0.221	1.00
GMRF	500	conditional	30	0.055	0.029	0.080	0.90
RF	500	marginal	30	0.055	0.027	0.082	0.87
RF + coordinates	500	marginal	30	0.065	0.040	0.091	0.90
GAM	500	marginal	30	0.080	0.032	0.129	0.67
GAM + s(X,Y)	500	marginal	18	0.058	0.006	0.111	0.78
GMRF	500	marginal	29	0.044	-0.011	0.099	0.52

Table S14. Optimism of random and spatial block cross-validation (cross-validation AUC minus held-out AUC) by model and cell: mean and standard error over replicates. Main text Table 7 pools models.

Model	n	Coverage	Target	Replicates	Random: mean	Random: SE	Block: mean	Block: SE
RF	30	clustered	conditional	30	0.017	0.031	0.018	0.030
RF	30	clustered	marginal	30	0.056	0.036	0.061	0.037
RF	30	dispersed	conditional	29	-0.104	0.030	-0.076	0.020
RF	30	dispersed	marginal	24	-0.026	0.030	-0.013	0.024
RF	50	clustered	conditional	30	-0.006	0.029	-0.008	0.028
RF	50	clustered	marginal	30	0.079	0.029	0.076	0.027
RF	50	dispersed	conditional	30	-0.015	0.018	-0.018	0.020
RF	50	dispersed	marginal	30	0.012	0.019	0.015	0.019
RF	100	clustered	conditional	30	0.034	0.018	0.038	0.016
RF	100	clustered	marginal	30	0.071	0.029	0.058	0.031
RF	100	dispersed	conditional	30	-0.006	0.013	-0.025	0.017
RF	100	dispersed	marginal	30	0.009	0.019	0.000	0.019
RF	200	clustered	conditional	30	0.014	0.017	0.018	0.016
RF	200	clustered	marginal	30	0.067	0.028	0.071	0.026
RF	200	dispersed	conditional	30	0.008	0.010	0.004	0.012
RF	200	dispersed	marginal	30	0.022	0.016	0.016	0.012
RF	500	clustered	conditional	30	0.044	0.020	0.047	0.017
RF	500	clustered	marginal	30	0.108	0.018	0.108	0.018
RF	500	dispersed	conditional	30	0.003	0.005	-0.007	0.004
RF	500	dispersed	marginal	30	0.039	0.009	0.043	0.008
RF + coordinates	30	clustered	conditional	30	0.022	0.032	0.016	0.029
RF + coordinates	30	clustered	marginal	30	0.071	0.032	0.049	0.042
RF + coordinates	30	dispersed	conditional	29	-0.107	0.034	-0.090	0.026
RF + coordinates	30	dispersed	marginal	24	-0.016	0.030	-0.026	0.031
RF + coordinates	50	clustered	conditional	30	0.012	0.032	0.012	0.030
RF + coordinates	50	clustered	marginal	30	0.066	0.032	0.067	0.031

Table S14 (continued)

Model	n	Coverage	Target	Replicates	Random: mean	Random: SE	Block: mean	Block: SE
RF + coordinates	50	dispersed	conditional	30	-0.041	0.024	-0.023	0.023
RF + coordinates	50	dispersed	marginal	30	0.008	0.027	0.014	0.026
RF + coordinates	100	clustered	conditional	30	0.061	0.018	0.073	0.018
RF + coordinates	100	clustered	marginal	30	0.082	0.031	0.068	0.029
RF + coordinates	100	dispersed	conditional	30	-0.004	0.012	-0.031	0.016
RF + coordinates	100	dispersed	marginal	30	0.033	0.019	0.019	0.019
RF + coordinates	200	clustered	conditional	30	0.046	0.015	0.048	0.015
RF + coordinates	200	clustered	marginal	30	0.094	0.031	0.090	0.029
RF + coordinates	200	dispersed	conditional	30	-0.006	0.010	-0.004	0.009
RF + coordinates	200	dispersed	marginal	30	0.042	0.018	0.044	0.015
RF + coordinates	500	clustered	conditional	30	0.075	0.022	0.071	0.018
RF + coordinates	500	clustered	marginal	30	0.117	0.017	0.109	0.017
RF + coordinates	500	dispersed	conditional	30	0.002	0.005	-0.007	0.005
RF + coordinates	500	dispersed	marginal	30	0.061	0.010	0.054	0.010
GAM	30	clustered	conditional	30	-0.054	0.039	-0.059	0.036
GAM	30	clustered	marginal	30	-0.019	0.048	-0.016	0.050
GAM	30	dispersed	conditional	29	-0.137	0.042	-0.104	0.033
GAM	30	dispersed	marginal	24	-0.068	0.039	-0.089	0.036
GAM	50	clustered	conditional	30	-0.048	0.030	-0.020	0.027
GAM	50	clustered	marginal	30	0.053	0.048	0.072	0.043
GAM	50	dispersed	conditional	30	-0.067	0.020	-0.051	0.028
GAM	50	dispersed	marginal	30	-0.023	0.026	-0.010	0.023
GAM	100	clustered	conditional	30	0.003	0.025	0.016	0.020
GAM	100	clustered	marginal	30	0.093	0.040	0.067	0.041
GAM	100	dispersed	conditional	30	-0.046	0.024	-0.051	0.026
GAM	100	dispersed	marginal	30	-0.004	0.023	-0.027	0.025
GAM	200	clustered	conditional	30	0.006	0.022	-0.000	0.020

Table S14 (continued)

Model	n	Coverage	Target	Replicates	Random: mean	Random: SE	Block: mean	Block: SE
GAM	200	clustered	marginal	30	0.082	0.034	0.077	0.032
GAM	200	dispersed	conditional	30	-0.010	0.010	-0.016	0.013
GAM	200	dispersed	marginal	30	0.015	0.021	0.027	0.014
GAM	500	clustered	conditional	30	0.022	0.024	0.026	0.020
GAM	500	clustered	marginal	30	0.133	0.028	0.131	0.028
GAM	500	dispersed	conditional	30	-0.005	0.004	-0.013	0.004
GAM	500	dispersed	marginal	30	0.058	0.008	0.055	0.008
GAM + s(X,Y)	30	clustered	conditional	30	0.045	0.030	0.054	0.036
GAM + s(X,Y)	30	clustered	marginal	12	0.018	0.063	0.033	0.062
GAM + s(X,Y)	30	dispersed	conditional	29	-0.063	0.035	-0.058	0.036
GAM + s(X,Y)	30	dispersed	marginal	22	0.038	0.038	-0.002	0.040
GAM + s(X,Y)	50	clustered	conditional	30	0.081	0.037	0.098	0.035
GAM + s(X,Y)	50	clustered	marginal	12	0.168	0.073	0.134	0.073
GAM + s(X,Y)	50	dispersed	conditional	30	-0.012	0.033	-0.027	0.029
GAM + s(X,Y)	50	dispersed	marginal	27	0.014	0.034	-0.023	0.036
GAM + s(X,Y)	100	clustered	conditional	30	0.091	0.024	0.085	0.023
GAM + s(X,Y)	100	clustered	marginal	20	0.037	0.047	0.007	0.042
GAM + s(X,Y)	100	dispersed	conditional	30	-0.041	0.018	-0.049	0.020
GAM + s(X,Y)	100	dispersed	marginal	29	0.045	0.026	0.024	0.024
GAM + s(X,Y)	200	clustered	conditional	30	0.091	0.030	0.085	0.026
GAM + s(X,Y)	200	clustered	marginal	20	0.105	0.046	0.118	0.052
GAM + s(X,Y)	200	dispersed	conditional	30	-0.036	0.015	-0.032	0.015
GAM + s(X,Y)	200	dispersed	marginal	29	0.044	0.026	0.062	0.018
GAM + s(X,Y)	500	clustered	conditional	30	0.118	0.037	0.129	0.033
GAM + s(X,Y)	500	clustered	marginal	18	0.158	0.033	0.149	0.032
GAM + s(X,Y)	500	dispersed	conditional	30	0.000	0.004	-0.006	0.004
GAM + s(X,Y)	500	dispersed	marginal	30	0.086	0.008	0.082	0.009

Table S14 (continued)

Model	n	Coverage	Target	Replicates	Random: mean	Random: SE	Block: mean	Block: SE
GMRF	30	clustered	conditional	29	-0.053	0.045	-0.062	0.041
GMRF	30	clustered	marginal	24	-0.014	0.052	0.000	0.051
GMRF	30	dispersed	conditional	19	-0.152	0.052	-0.099	0.045
GMRF	30	dispersed	marginal	18	-0.052	0.043	-0.050	0.042
GMRF	50	clustered	conditional	30	0.005	0.032	0.022	0.024
GMRF	50	clustered	marginal	27	0.062	0.049	0.087	0.048
GMRF	50	dispersed	conditional	12	-0.044	0.052	-0.016	0.028
GMRF	50	dispersed	marginal	22	0.007	0.045	0.020	0.035
GMRF	100	clustered	conditional	26	0.010	0.025	0.020	0.020
GMRF	100	clustered	marginal	30	0.125	0.040	0.115	0.042
GMRF	100	dispersed	conditional	22	-0.020	0.019	-0.001	0.021
GMRF	100	dispersed	marginal	21	0.050	0.026	0.052	0.024
GMRF	200	clustered	conditional	30	0.005	0.022	0.004	0.020
GMRF	200	clustered	marginal	29	0.061	0.030	0.051	0.029
GMRF	200	dispersed	conditional	25	-0.012	0.014	-0.023	0.016
GMRF	200	dispersed	marginal	27	0.082	0.013	0.080	0.015
GMRF	500	clustered	conditional	30	0.024	0.028	0.040	0.020
GMRF	500	clustered	marginal	30	0.122	0.027	0.128	0.027
GMRF	500	dispersed	conditional	30	-0.004	0.003	-0.012	0.004
GMRF	500	dispersed	marginal	29	0.100	0.012	0.097	0.011

Table S15a. Fit failure rates by model, sample size and coverage: proportion of replicates with an error, with no finite prediction or non-convergence, with a degenerate prediction surface, and unusable on any of these grounds.

Model	n	Coverage	Runs	Error	No prediction	Degenerate	Unusable
RF	30	dispersed	53	0.000	0.000	0.000	0.000
RF	30	clustered	60	0.000	0.000	0.000	0.000
RF	50	dispersed	60	0.000	0.000	0.000	0.000
RF	50	clustered	60	0.000	0.000	0.000	0.000
RF	100	dispersed	60	0.000	0.000	0.000	0.000
RF	100	clustered	60	0.000	0.000	0.000	0.000
RF	200	dispersed	60	0.000	0.000	0.000	0.000
RF	200	clustered	60	0.000	0.000	0.000	0.000
RF	500	dispersed	60	0.000	0.000	0.000	0.000
RF	500	clustered	60	0.000	0.000	0.000	0.000
RF + coordinates	30	dispersed	53	0.000	0.000	0.000	0.000
RF + coordinates	30	clustered	60	0.000	0.000	0.000	0.000
RF + coordinates	50	dispersed	60	0.000	0.000	0.000	0.000
RF + coordinates	50	clustered	60	0.000	0.000	0.000	0.000
RF + coordinates	100	dispersed	60	0.000	0.000	0.000	0.000
RF + coordinates	100	clustered	60	0.000	0.000	0.000	0.000
RF + coordinates	200	dispersed	60	0.000	0.000	0.000	0.000
RF + coordinates	200	clustered	60	0.000	0.000	0.000	0.000
RF + coordinates	500	dispersed	60	0.000	0.000	0.000	0.000
RF + coordinates	500	clustered	60	0.000	0.000	0.000	0.000
GAM	30	dispersed	53	0.000	0.000	0.000	0.000
GAM	30	clustered	60	0.000	0.000	0.000	0.000
GAM	50	dispersed	60	0.000	0.000	0.000	0.000
GAM	50	clustered	60	0.000	0.000	0.000	0.000
GAM	100	dispersed	60	0.000	0.000	0.000	0.000
GAM	100	clustered	60	0.000	0.000	0.000	0.000

Table S15a (continued)

Model	n	Coverage	Runs	Error	No prediction	Degenerate	Unusable
GAM	200	dispersed	60	0.000	0.000	0.000	0.000
GAM	200	clustered	60	0.000	0.000	0.000	0.000
GAM	500	dispersed	60	0.000	0.000	0.000	0.000
GAM	500	clustered	60	0.000	0.000	0.000	0.000
GAM + s(X,Y)	30	dispersed	53	0.000	0.000	0.038	0.038
GAM + s(X,Y)	30	clustered	60	0.000	0.000	0.300	0.300
GAM + s(X,Y)	50	dispersed	60	0.000	0.000	0.050	0.050
GAM + s(X,Y)	50	clustered	60	0.000	0.000	0.300	0.300
GAM + s(X,Y)	100	dispersed	60	0.000	0.000	0.017	0.017
GAM + s(X,Y)	100	clustered	60	0.000	0.000	0.167	0.167
GAM + s(X,Y)	200	dispersed	60	0.000	0.000	0.017	0.017
GAM + s(X,Y)	200	clustered	60	0.000	0.000	0.167	0.167
GAM + s(X,Y)	500	dispersed	60	0.000	0.000	0.000	0.000
GAM + s(X,Y)	500	clustered	60	0.000	0.000	0.200	0.200
GMRF	30	dispersed	53	0.000	1.000	0.000	0.302
GMRF	30	clustered	60	0.000	0.867	0.000	0.117
GMRF	50	dispersed	60	0.000	0.967	0.000	0.417
GMRF	50	clustered	60	0.000	0.800	0.000	0.050
GMRF	100	dispersed	60	0.000	0.817	0.000	0.267
GMRF	100	clustered	60	0.000	0.750	0.000	0.067
GMRF	200	dispersed	60	0.000	0.933	0.000	0.133
GMRF	200	clustered	60	0.000	0.533	0.000	0.017
GMRF	500	dispersed	60	0.000	0.467	0.000	0.017
GMRF	500	clustered	60	0.000	0.450	0.000	0.000

Table S15b. The sdmTMB sanity flag: numbers of usable random field fits failing and passing the combined flag (positive definite Hessian, optimiser convergence, small gradients) by sample size and coverage, with their mean held-out AUC. The flag was treated as a diagnostic rather than an exclusion criterion.

n	Coverage	Usable fits failing the flag	Usable fits passing	AUC, failing	AUC, passing
30	dispersed	37	NA	0.777	NA
30	clustered	45	8	0.710	0.773
50	dispersed	33	2	0.782	0.829
50	clustered	45	12	0.711	0.711
100	dispersed	33	11	0.804	0.815
100	clustered	41	15	0.712	0.722
200	dispersed	48	4	0.803	0.844
200	clustered	31	28	0.790	0.780
500	dispersed	27	32	0.815	0.839
500	clustered	27	33	0.766	0.788

Table S16a. Mixed model contrasts: estimates, standard errors and p values from the linear mixed model for the four within-family and algorithm contrasts by cell (emmeans, no multiplicity adjustment).

Contrast	n	Coverage	Target	Estimate	SE	p value
rf_spatial - rf	30	dispersed	conditional	0.017	0.025	0.5
gam_spatial - gam	30	dispersed	conditional	-0.087	0.025	0.00059
gmrf - gam_spatial	30	dispersed	conditional	0.107	0.028	0.00017
rf - gam	30	dispersed	conditional	0.037	0.025	0.14
rf_spatial - rf	50	dispersed	conditional	0.022	0.025	0.39
gam_spatial - gam	50	dispersed	conditional	-0.094	0.025	0.00016
gmrf - gam_spatial	50	dispersed	conditional	0.121	0.032	0.00015
rf - gam	50	dispersed	conditional	0.012	0.025	0.62
rf_spatial - rf	100	dispersed	conditional	0.031	0.025	0.21
gam_spatial - gam	100	dispersed	conditional	-0.060	0.025	0.015
gmrf - gam_spatial	100	dispersed	conditional	0.064	0.027	0.017
rf - gam	100	dispersed	conditional	-0.012	0.025	0.62
rf_spatial - rf	200	dispersed	conditional	0.028	0.025	0.25
gam_spatial - gam	200	dispersed	conditional	0.004	0.025	0.88
gmrf - gam_spatial	200	dispersed	conditional	-0.007	0.026	0.78
rf - gam	200	dispersed	conditional	-0.022	0.025	0.38
rf_spatial - rf	500	dispersed	conditional	0.031	0.025	0.21
gam_spatial - gam	500	dispersed	conditional	0.014	0.025	0.58
gmrf - gam_spatial	500	dispersed	conditional	-0.009	0.025	0.7
rf - gam	500	dispersed	conditional	-0.017	0.025	0.5
rf_spatial - rf	30	clustered	conditional	-0.040	0.025	0.11
gam_spatial - gam	30	clustered	conditional	-0.135	0.025	5.8e-08
gmrf - gam_spatial	30	clustered	conditional	0.144	0.025	1e-08
rf - gam	30	clustered	conditional	0.032	0.025	0.19
rf_spatial - rf	50	clustered	conditional	-0.020	0.025	0.42
gam_spatial - gam	50	clustered	conditional	-0.151	0.025	1.4e-09

Table S16a (continued)

Contrast	n	Coverage	Target	Estimate	SE	p value
gmrf - gam_spatial	50	clustered	conditional	0.138	0.025	2.7e-08
rf - gam	50	clustered	conditional	0.018	0.025	0.48
rf_spatial - rf	100	clustered	conditional	-0.026	0.025	0.29
gam_spatial - gam	100	clustered	conditional	-0.138	0.025	3.1e-08
gmrf - gam_spatial	100	clustered	conditional	0.115	0.026	8.6e-06
rf - gam	100	clustered	conditional	0.004	0.025	0.88
rf_spatial - rf	200	clustered	conditional	-0.019	0.025	0.45
gam_spatial - gam	200	clustered	conditional	-0.102	0.025	4.2e-05
gmrf - gam_spatial	200	clustered	conditional	0.096	0.025	0.00011
rf - gam	200	clustered	conditional	-0.021	0.025	0.4
rf_spatial - rf	500	clustered	conditional	0.001	0.025	0.97
gam_spatial - gam	500	clustered	conditional	-0.101	0.025	5.2e-05
gmrf - gam_spatial	500	clustered	conditional	0.098	0.025	8.2e-05
rf - gam	500	clustered	conditional	-0.031	0.025	0.21
rf_spatial - rf	30	dispersed	marginal	0.023	0.028	0.42
gam_spatial - gam	30	dispersed	marginal	-0.091	0.028	0.0014
gmrf - gam_spatial	30	dispersed	marginal	0.115	0.031	0.00018
rf - gam	30	dispersed	marginal	0.045	0.028	0.1
rf_spatial - rf	50	dispersed	marginal	0.004	0.025	0.88
gam_spatial - gam	50	dispersed	marginal	-0.070	0.026	0.0058
gmrf - gam_spatial	50	dispersed	marginal	0.084	0.028	0.0023
rf - gam	50	dispersed	marginal	0.031	0.025	0.21
rf_spatial - rf	100	dispersed	marginal	0.017	0.025	0.5
gam_spatial - gam	100	dispersed	marginal	-0.104	0.025	3.6e-05
gmrf - gam_spatial	100	dispersed	marginal	0.080	0.028	0.0038
rf - gam	100	dispersed	marginal	-0.011	0.025	0.67
rf_spatial - rf	200	dispersed	marginal	0.013	0.025	0.6

Table S16a (continued)

Contrast	n	Coverage	Target	Estimate	SE	p value
gam_spatial - gam	200	dispersed	marginal	-0.043	0.025	0.09
gmrf - gam_spatial	200	dispersed	marginal	0.003	0.026	0.89
rf - gam	200	dispersed	marginal	-0.012	0.025	0.64
rf_spatial - rf	500	dispersed	marginal	0.012	0.025	0.64
gam_spatial - gam	500	dispersed	marginal	-0.013	0.025	0.6
gmrf - gam_spatial	500	dispersed	marginal	-0.022	0.025	0.38
rf - gam	500	dispersed	marginal	-0.007	0.025	0.78
rf_spatial - rf	30	clustered	marginal	0.000	0.025	0.99
gam_spatial - gam	30	clustered	marginal	-0.066	0.033	0.045
gmrf - gam_spatial	30	clustered	marginal	0.084	0.034	0.014
rf - gam	30	clustered	marginal	0.048	0.025	0.051
rf_spatial - rf	50	clustered	marginal	0.005	0.025	0.84
gam_spatial - gam	50	clustered	marginal	-0.115	0.033	0.00049
gmrf - gam_spatial	50	clustered	marginal	0.110	0.033	0.00099
rf - gam	50	clustered	marginal	0.050	0.025	0.046
rf_spatial - rf	100	clustered	marginal	0.004	0.025	0.86
gam_spatial - gam	100	clustered	marginal	-0.032	0.028	0.24
gmrf - gam_spatial	100	clustered	marginal	-0.012	0.028	0.67
rf - gam	100	clustered	marginal	0.024	0.025	0.34
rf_spatial - rf	200	clustered	marginal	-0.009	0.025	0.71
gam_spatial - gam	200	clustered	marginal	-0.047	0.028	0.092
gmrf - gam_spatial	200	clustered	marginal	0.056	0.028	0.047
rf - gam	200	clustered	marginal	0.020	0.025	0.42
rf_spatial - rf	500	clustered	marginal	0.001	0.025	0.97
gam_spatial - gam	500	clustered	marginal	-0.008	0.029	0.77
gmrf - gam_spatial	500	clustered	marginal	0.010	0.029	0.73

rf - gam	500	clustered	marginal	0.019	0.025	0.45
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Table S16b. Variance components of the mixed model for held-out AUC.

Component	Variance	Standard deviation
sp_rep	0.00012	0.011
species	0.00170	0.041
Residual	0.00925	0.096

Supplementary references

References cited only in this Supporting Information; all others are in the main text reference list.

Allouche, O., Tsoar, A., & Kadmon, R. (2006). Assessing the accuracy of species distribution models: Prevalence, kappa and the true skill statistic (TSS). *Journal of Applied Ecology*, 43(6), 1223–1232.

Bahn, V., & McGill, B. J. (2013). Testing the predictive performance of distribution models. *Oikos*, 122(3), 321–331.

Behrens, T., Schmidt, K., Viscarra Rossel, R. A., Gries, P., Scholten, T., & MacMillan, R. A. (2018). Spatial modelling with Euclidean distance fields and machine learning. *European Journal of Soil Science*, 69(5), 757–770.

Bell, R. J., Richardson, D. E., Hare, J. A., et al. (2014). Disentangling the effects of climate, abundance, and size on the distribution of marine fish: An example based on four stocks from the Northeast US shelf. *ICES Journal of Marine Science*, 72(5), 1311–1322.

Flynn, A. J., & Marshall, N. J. (2013). Lanternfish (Myctophidae) zoogeography off eastern Australia: A comparison with physicochemical biogeography. *PLOS ONE*, 8(12), e80950.

Kristensen, K., Nielsen, A., Berg, C. W., et al. (2016). TMB: Automatic differentiation and Laplace approximation. *Journal of Statistical Software*, 70(5), 1–21.

Ovaskainen, O., & Abrego, N. (2020). *Joint Species Distribution Modelling: With Applications in R*. Cambridge University Press.

Pichler, M., & Hartig, F. (2021). A new joint species distribution model for faster and more accurate inference of species associations from big community data. *Methods in Ecology and Evolution*, 12(11), 2159–2173.

Redfern, J. V., Ferguson, M. C., Becker, E. A., et al. (2006). Techniques for cetacean habitat modeling. *Marine Ecology Progress Series*, 310, 271–295.

Rooper, C. N., Sigler, M. F., Goddard, P., et al. (2016). Validation and improvement of species distribution models for structure-forming invertebrates in the eastern Bering Sea with an independent survey. *Marine Ecology Progress Series*, 551, 117–130.

Rooper, C. N., Wilborn, R., Goddard, P., et al. (2018). Validation of deep-sea coral and sponge distribution models in the Aleutian Islands, Alaska. *ICES Journal of Marine Science*, 75(1),

199–209.

Rooper, C. N., Williams, K., Towler, R., et al. (2025). Lessons learned from testing the predictions of species distribution models for deep-sea corals and sponges in the Gulf of Alaska with comparisons to other Alaska ecosystems. *ICES Journal of Marine Science*, 82(10).

Zurell, D., Franklin, J., König, C., et al. (2020). A standard protocol for reporting species distribution models. *Ecography*, 43(9), 1261–1277.